Surname [illegible] *Van*	**Initials** M ***Voorletters*** **Stud. No.** [illegible]
The undersigned hereby confirms that they are familiar with and accepts the examination regulations of the University of Pretoria. Your cheat sheet must be handed in with this question paper.	*Die ondergetekende bevestig hiermee dat hulle bewus is van die eksamenregulasies van die Universiteit van Pretoria en dat hy/sy hom/haar daaraan onderwerp. Jy moet jou formuleblad inhandig met die vraestel.*

15

Signature [redacted] *Handtekening* *Date* 14 Feb *Datum*

Class Test 1 **14 / 02 / 2013** ***Klastoets 1***

FSK116

Total/Totaal: 30

Question 1/*Vraag 1*

Calculate the following using your calculator and fill in your answer in the space provided:

Maak gebruik van jou sakrekenaar om die volgende te bereken en skryf jou antwoord in die spasie langsaan:

[4]

$\cos(2.345\pi \text{ rad})$ = cos(422.1°)	0.47
$\log\left(\sqrt{12.67}\right)$	0.55
$\ln(12.67)$	2.54
$e^{4.25}$	70.11

4

Convert the following and fill in your answer in the space provided:

Maak die volgende omskakelings en skryf jou antwoord in die spasie langsaan:

[3]

0.567 rad to degrees 0.567 *rad na grade*	32.49°
216° to radians 216° *na radiaal*	$\frac{6}{5}\pi$ = 3.77 rad
$4\pi/3$ rad to degrees $4\pi/3$ *rad na grade*	240°

3

Question 2/*Vraag 2*

2.1 Determine the derivative of the function $y = y(x) = \sin(2x)\cos(2x)$ and calculate the gradient of the function at $x = \pi/3$ rad.

Bepaal die afgeleide van die funksie $y = y(x) = \sin(2x)\cos(2x)$ *en bereken die helling van die grafiek by* $x = \pi/3$ *rad.*

[4]

$dy/dx = -(2)(\cos 2x)(2)(\sin 2x)$

$\pi/3$ rad = 60°

$\therefore dy/dx = -(2)(\cos 2(60))(2)(\sin 2(60))$

$= \sqrt{3} = 1.73$

THE GRADIENT OF THE FUNCTION AT $x = \pi/3$ rad IS 1.73

7

Question 3/*Vraag 3*

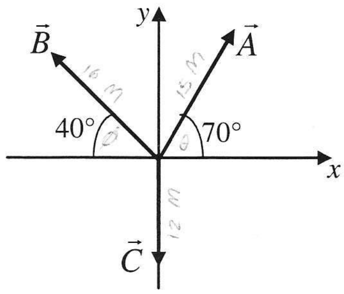
3.1 Determine the indefinite integral of
Bereken die onbepaalde integraal van

[2]

3.2 Determine the definite integral of
Bereken die bepaalde integraal van

[3]

Question 4/*Vraag 4*



Consider the vectors in the sketch. Vector $\vec{A}$ has a magnitude of 15 m, vector $\vec{B}$ has a magnitude of 16 m and vector $\vec{C}$ has a magnitude of 12 m as shown in the sketch.

Beskou die vektore in die skets. Vektor $\vec{A}$ het 'n grootte van 15 m, vektor $\vec{B}$ het 'n grootte van 16 m en vektor $\vec{C}$ het 'n grootte van 12 m, soos aangedui in die skets.

a. Write the three vectors using unit vector notation.
Skryf die drie vektore in eenheids-vektor notasie.

[3]

$a_x = a\cos\theta = 5.13$ m $\quad a_y = a\sin\theta = 14.1$ m

$\therefore \vec{A} = (5.13\ m)\hat{i} + (14.1\ m)\hat{j}$

$b_x = b\cos(180-\phi) = \cancel{10.28\ m}$ -12.26 m $\quad b_y = b\sin(180-\phi) = 10.28$ m

$\therefore \vec{B} = (-12.26\ m)\hat{i} + (10.28\ m)\hat{j}$

$\vec{C} = (0\ m)\hat{i} + (-12.0\ m)\hat{j} = (-12.0\ m)\hat{j}$

3

3/3

b. Determine the magnitude and direction of the resulting vector $\vec{r} = \vec{A} + \vec{B} - \vec{C}$ using the component method.

Bereken die grootte en rigting van die vektor $\vec{r} = \vec{A} + \vec{B} - \vec{C}$ deur gebruik te maak van die komponent-metode.

[5]

$\vec{r} = \vec{A} + \vec{B} - \vec{C}$

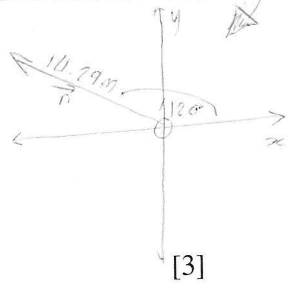
$\vec{r} = (a_x + b_x - c_x)\hat{i} + (a_y + b_y - c_y)\hat{j}$

$= [(5.13\,m) + (-12.26\,m) - (0\,m)]\hat{i} + [(14.1\,m) + (10.28\,m) - (12\,m)]\hat{j}$

$= (-7.13\,M)\hat{i} + (12.38\,M)\hat{j}$

$\vec{r}_{MAGNITUDE}$: $\vec{r} = \sqrt{(r_x)^2 + (r_y)^2} = \sqrt{(-7.13\,m)^2 + (12.38\,m)^2} = 14.29\,M$

$\vec{r}_{ANGLE}$: $\tan\theta = \frac{r_y}{r_x} = \left(\frac{12.38}{-7.13}\right) = -60.06° \therefore 180 - 60.06 = 120°$



Question 5/*Vraag 5*

Two vectors are given by $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Determine the following:

Twee vektore word gegee as $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ en $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Bepaal die volgende:

(a) $\vec{a} \times \vec{b}$

$a_x = 3.0$ $b_x = 2.0$
$a_y = 5.0$ $b_y = 4.0$
$a_z = 0$ $b_z = 0$

[3]

$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z)\hat{i} + (a_z b_x + b_z a_x)\hat{j} + (a_x b_y - b_x a_y)\hat{k}$

$= [(5)(0) - (4)(0)]\hat{i} + [(0)(2) + (0)(3)]\hat{j} + [(3)(4) - (2)(5)]\hat{k}$

$= 0\hat{i} + 0\hat{j} + 2.0\hat{k} = 2.0\hat{k}$

(b) $\vec{a} \cdot \vec{b}$

[3]

$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$

$= (3)(2)\hat{i} + (5)(4)\hat{j} + (0)(0)\hat{k}$

$= 6\hat{i} + 20\hat{j} + 0\hat{k} = (6.0)\hat{i} + (20.0)\hat{j}$

THE END / DIE EINDE

THANK YOU

~~$a_x = 5.13$~~
~~$a_y = 14.1$~~
~~$b_x = -12.26$~~
~~$b_y = 10.28$~~
~~$c_x = 0$~~
~~$c_y = 12$~~